Jerby-Arnon 2018 melanoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

θ derived by connectivity (θ=2 fragments the manifold into 2 components / 185 detached; θ=5 = one component, 0 detached, stable through θ=10/20). Per-patient CNV texture is θ-invariant (kNN-purity 2–8× at every θ) but no patient forms a disconnected island. Decisive: each of the 8 patients TRAVERSES the shared embryonic axis (ρ(axis,Crest) −0.66..−0.95, spanning 59–96%) — one continuous gastrulation→neural-crest gradient, not a per-patient archipelago. iLISI 1.00→3.71 — mixed & continuous; no island removed.

Gate 2 — module geography → GO (non-branching)

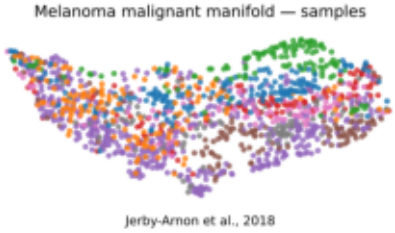
LOW-pSom, Crest-dominant melanoma (raw means Gast −0.49, pSom −0.14, Crest +0.20). Gast & pSom OVERLAP at the origin (spatial r=+0.93; joint-high enrichment 1.44) and decline together along pseudotime (Gast ρ=−0.98, pSom ρ=−0.92), opposed to a terminal neural-crest end that rises (Crest ρ=+0.93; decile pt=0.97). Crest anti-correlated in space (Crest↔Gast −0.91, Crest↔pSom −0.82); joint pSom&Crest-high depleted (0.28) — inverse gradient, not a hub. Crest peaks at the terminus, not a Gast arm — no inversion flag.

Gate 3 — cross-repression → NA

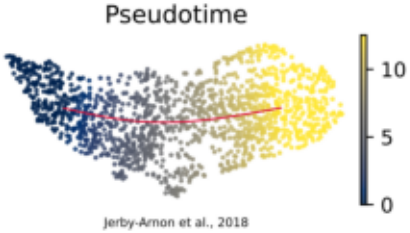
Single monotone axis — no bifurcation (curve: 2 tips, 0 forks; PPT tree yields only spurious forks) and no Gast/pSom-overlap Mixed hub, so there is no branch pair to test for Gast↔pSom cross-repression.

STEP FIGURES

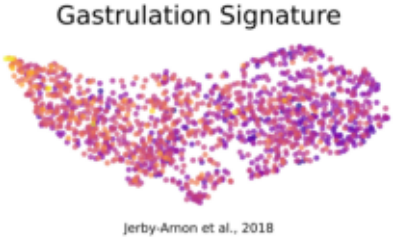
Manifold — samples



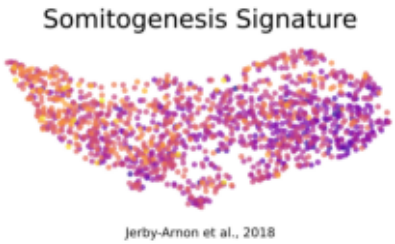
Pseudotime



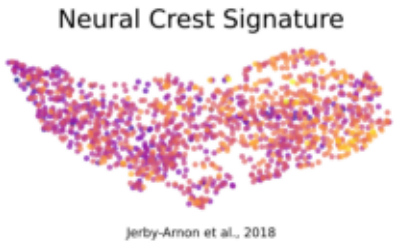
Gastrulation



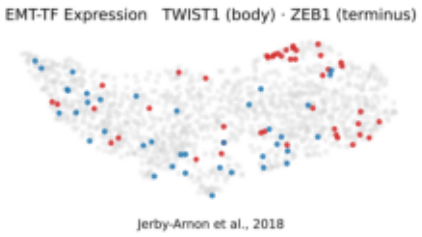
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

Cutaneous/metastatic melanoma (Jerby-Arnon et al., 2018), SmartSeq2 — 23 samples / 7,186 cells (master scope); malignant only: 2,018 cells; ≥100-cell floor → 1,379 cells (8 samples) on the trajectory (no island removal, no scoping).

Verdict.

Non-branching pattern reproduces — G1 GO, G2 GO, G3 NA. The melanoma malignant manifold is a single monotone gastrulation/somitogenesis → terminal-neural-crest gradient (Gast ρ=−0.98, pSom ρ=−0.92, Crest ρ=+0.93); it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test (Linear class). The atlas's LOW-pSom, Crest-dominant case — Crest the only positive module (+0.20), Gast (−0.49) and pSom (−0.14) globally weak, the neural-crest identity of a neural-crest-derived tumour. Two decisive derived calls: θ=5 by connectivity, and a patient-SHARED continuous-axis test clearing Gate 1 despite θ-invariant per-patient CNV texture. EMT terminus-loaded: ZEB1 marks the neural-crest terminus, TWIST1 the body factor (contrast Puram's SNAI1/ZEB1).

Parameter choices

Cells / patients	1,379 / 8 (≥100 floor, 0 islands)	**Integration**	Harmony θ=5 (connectivity); PCA50
Island removed	none (patient-shared axis)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	Gast/pSom-high, Crest-low tip	**Geometry**	Linear (non-branching); 2 tips / 0 forks
EMT-TFs	TWIST1 (body) · ZEB1 (terminus); terminus-loaded, no root TF		